

Abstract

Lightweight mortars for precast ferrocement construction face an inherent engineering trade-off: reducing density with Expanded Polystyrene (EPS) beads severely compromises compressive strength and toughness, driven by poor interfacial bonding and buoyancy-induced segregation. This research resolves that paradox through a ternary synergistic system surface-coated EPS for density reduction, ball-milled Rice Husk Ash (RHA) for pozzolanic matrix densification, and Polypropylene (PP) micro-fibers for crack-bridging toughness systematically optimized via Response Surface Methodology with a Face-Centered Central Composite Design (CCF). The resulting matrix achieves a density of 1993 kg/m³ (13% lighter than the plain control) while retaining a compressive strength of 15.62 MPa, flexural strength of 3.59 MPa, and toughness of 18.24 mJ/mm³, with apparent porosity essentially unchanged from the dense baseline. These results confirm that a properly engineered ternary matrix can decouple density reduction from strength loss, offering a high-performance, viable lightweight solution for non-load-bearing and also load bearing precast partition panels.

Introduction

The Paradox

Conventional cement mortar runs 2200–2400 kg/m³ heavy enough to demand high-capacity lifting equipment, increase handling stress during demolding, and risk early cracking before precast panels reach full strength. EPS beads are the obvious fix for weight, but raw EPS is a smooth, hydrophobic void that wrecks the very strength and toughness the mortar needs to survive that handling.

Microstructural Failure

Uncoated EPS creates a weak, calcium-rich Interfacial Transition Zone (ITZ) around every bead essentially a network of pre-formed failure planes. Its low density also causes it to float and segregate during casting, compounding the problem before the mortar even sets.

The Objective

Engineer a ternary system that neutralizes all three failure modes at once: heal the ITZ chemically, lock the EPS mechanically into the paste, and arrest the resulting brittleness without exceeding a target density of 2000 kg/m³.

Methodology

Mix optimization was carried out using Response Surface Methodology (RSM) with a Face-Centered Central Composite Design (CCF), allowing each mechanism to be isolated and statistically validated before combining them into the final ternary formulation.

Phase 1: Chemical Healing (Ball-Milled RHA)

Rice Husk Ash was ball-milled to a reactive pozzolanic fineness, driving a secondary hydration reaction that consumes weak Calcium Hydroxide and deposits dense Calcium Silicate Hydrate (C-S-H) gel directly at the aggregate–paste boundary.



Figure 1 : Processed RHA powder

Phase 2: Physical Interlocking (Coated EPS)

EPS beads were surface-coated using a Polyvinyl Acetate (PVAc) and M-sand matrix. This neutralized buoyancy (preventing segregation) and provided a roughened surface for mechanical interlocking with the cement paste.



Figure 2 : Coated EPS beads vs RAW EPS beads

Phase 3: Mechanical Toughening (PP Micro-Fibers)

Polypropylene micro-fibers were dosed incrementally (0–2.0% by volume) into the optimized RHA–coated-EPS matrix to bridge micro-cracks and extend post-peak ductility, up to the threshold where geometric "fiber balling" begins to reverse the gains.



Figure 3 : Loose PP fibers

Result and Discussion

A. Interfacial Transition Zone (ITZ) Densification

SEM-EDS diagnostic mapping mathematically confirms the successful pozzolanic reaction of RHA: the elemental Calcium-to-Silicon (Ca/Si) atomic ratio at the ITZ dropped from 24.85 to 5.65 a ~77% reduction confirming the transformation of a weak, crystalline Portlandite-rich boundary into a dense, amorphous, load-bearing C-S-H gel.

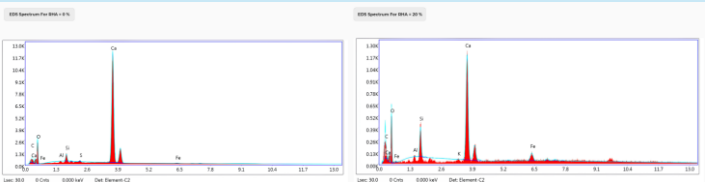


Figure 4 : EDS Spectra comparing the Ca/Si ratio drop from 24.85 to 5.65]

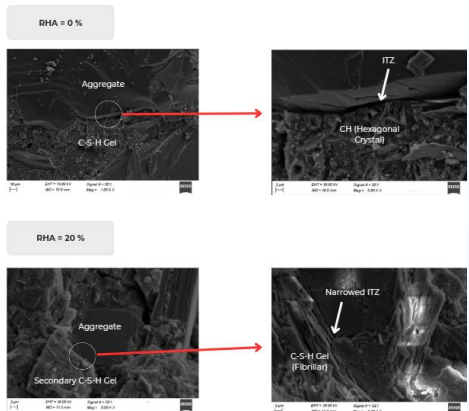


Figure 5 : SEM Micrographs (Without RHA clear ITZ vs. with RHA healed ITZ)

B. Physical Coating & Compressive Recovery

Surface coating eliminated the classic EPS "wall effect." At 40% EPS replacement, coating recovered compressive strength from 4.20 to 7.15 MPa a 70.24% recovery proof that mechanical interlocking, not just chemistry, is essential to making EPS mortar viable.

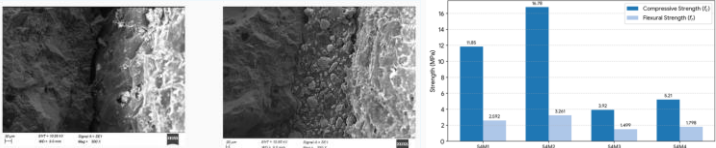


Figure 6 : SEM Micrographs and strength comparison

C. Fiber Optimization & The Degradation Plateau

On the optimized 10% RHA + 20% coated-EPS matrix, 0.5% PP fiber emerged as the practical optimum lifting toughness from 14.71 to 18.24 mJ/mm³ (+24%) and flexural strength from 3.26 to 3.59 MPa, while sacrificing only ~7% compressive strength. Pushing dosage to an extreme 2.0% confirmed this isn't a soft ceiling: compressive strength continued declining to 11.06 MPa and flexural strength fell below the unreinforced baseline, mathematically confirming a genuine degradation plateau rather than an artifact of a narrow test window.

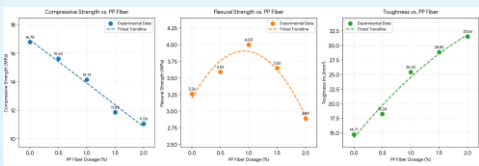


Figure 7 : The line graphs showing the steep downward trendline of strength vs. fiber dosage

Conclusions

1.Optimized Output:

The finalized ternary composite 10% RHA, 20% coated EPS, 0.5% PP fiber achieves a lightweight density of 1993 kg/m³ (13% lighter than the plain control) while retaining 15.62 MPa compressive strength, 3.59 MPa flexural strength, and 18.24 mJ/mm³ toughness, with porosity essentially unchanged from the dense baseline.

2.Scientific Validation:

Macro-mechanical testing and micro-analytical SEM–EDS diagnostics tell the same story from two independent directions porosity refinement and ITZ densification are the true strength-controlling mechanisms, not incidental correlations.

3.Real-World Viability:

This ternary design methodology is generalizable and cost-effective, offering a practical route to lighter, more transportable precast ferrocement wall panels without the strength penalty that has historically limited EPS mortar adoption.

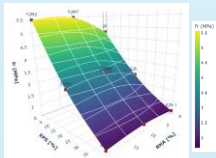


Figure 8 : Flexural strength variation

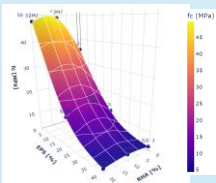


Figure 9 : Compressive strength variation

References

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